Mawlana Bhashani Science and Technology University

Department of Information and Communication Technology

4th year 1st Semester B. Sc (Engg.) Final Examination 2024

Course Title: Digital Image Processing — Course Code: ICT 4111

Full Marks: 70 — Time: 3 hour

Answer any five from the following questions

1. a) Explain about image sampling and quantization process with proper steps. **4**
   b) Briefly describe the fundamental steps in digital image processing which can be applied to images? **6**
   c) List out the applications of digital image processing in modern era. **4**

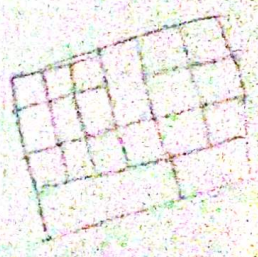
2. a) Illustrate about the neighboring, 4-adjacency, 8-adjacency and m-adjacency of a pixels with suitable examples. **4**
   b) Let V={0, 1}, Compute Euclidian distance, city-block-distance and chessboard distance between p and q for the given image. **3**

$$A=\begin{bmatrix} 1(p) & 1 & 2 & 3 \\ 0 & 2 & 2 & 1 \\ 1 & 1 & 0 & 2(q) \\ 2 & 1 & 2 & 1 \end{bmatrix}.$$

   c) What do you mean by image sensing and acquisition? Briefly explain how can image formation in the eye. **5**
   d) What is the difference between image enhancement and image restoration? **2**

3. a) Define Histogram of a digital image. Explain how histogram is useful in image enhancement. **4**
   b) A 3-bit image of size 64 × 64 pixel has the following histogram: **3**

Intensity Levels	Frequency
0	790
1	1023
2	850
3	656
4	329
5	245
6	122
7	81



Tabulate the histogram of the equalized image using the transform function that you proposed.

   c) i) A pixel is given in RGB color format with R=50, G=0.55, B=35. Convert it to HIS color format. **4**
   ii) The HIS values of pixel are H=0.75. S=0.55, I=80. Determine its RGB value.
   d) How can we distinguish dark, bright, low contrast and high contrast images from histograms? **3**

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4. a) What is unsharp mask? Write down the steps of unsharp masking procedure? **4**

b) Explain the noises are used for image restoration with their expressions and probability density function **6**

c) Apply Laplacian filter of the given image A on the center pixel. **4**

$$A = \begin{bmatrix} 8 & 5 & 4 \\ 0 & 6 & 2 \\ 5 & 8 & 9 \end{bmatrix}$$

5. a) Write the expression for Harmonic and contra-harmonic mean filter with their importance. **4**

b) Convert the given spatial domain image using Fourier Transform and Perform ideal high-pass filter to smoother the image. Choose D_o is 0.8. Show the step by step procedure for the following image. **10**

$$f(m,n) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

6. a) Give a formal definition of image segmentation. Describe the reason splitting and margin procedure for image segmentation. **5**

b) Narrate how motion can be used in image segmentation. **3**

c) Explain k-means clustering algorithm for image segmentation. **3**

d) Use the LZW coding algorithm to encode the 7-bit ASCII string "aaaaaaaaaa". **3**

7. a) What do you mean by image compression 10:1? **2**

b) Construct the Huffman codes for the given image is of size 8 × 8 and the gray level values of such pixels are shown in the matrix form as below. Compute the entropy. **8**

$$\begin{bmatrix} 12 & 12 & 12 & 12 & 12 & 12 & 12 & 12 \\ 12 & 12 & 12 & 12 & 12 & 12 & 12 & 12 \\ 12 & 12 & 12 & 12 & 12 & 12 & 12 & 12 \\ 12 & 12 & 200 & 200 & 200 & 100 & 100 & 100 \\ 50 & 50 & 200 & 200 & 200 & 100 & 100 & 100 \\ 50 & 50 & 200 & 200 & 200 & 100 & 100 & 100 \\ 50 & 50 & 20 & 20 & 20 & 20 & 5 & 5 \\ 50 & 50 & 20 & 20 & 20 & 5 & 5 & 5 \end{bmatrix}$$

c) What is Image watermarking? Briefly explain the classifications of image watermarking. **4**

8. a) Briefly explain erosion and dilation with examples. **4**

b) Use the graph theoretic approach to find the edge corresponding to the minimum cost path for the sub-image shown in Figure below: **5**

$$\begin{bmatrix} 7 & 5 & 2 \\ 8 & 1 & 3 \\ 6 & 3 & 5 \end{bmatrix}$$

c) Describe Hit and Miss transformation technique in morphological image processing. **5**

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